Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/01/2007
PI Name:	Roby, Richard Ph.D.		
Project Title:	Development of an Engineering Tool for the Determination of Suppression Device Placement in Reduced Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21045	**Congressional District:**	7
Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhang , Wei (Combustion Science & Engineering, Inc.) Klassen , Michael (Combustion Science & Engineering, Inc.) Olenick , Stephen (Combustion Science & Engineering, Inc.)		
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Task Description:	Prediction of the optimum size and location of suppression systems for fire extinguishment in reduced gravity facilities requires either experimental measurements or computer simulations. The experimental approach requires reduced gravity testing facilities. Although this approach can generate reliable data, conducting experiments in reduced gravity is expensive, time-consuming, and only practical on a reduced-scale basis. The current experimental data available in the literature is not suitable for full-scale estimation of fire and smoke motion in space facilities, especially for determining placement of automatic suppression devices. An alternative solution to designing these systems is the use of computer modeling techniques to predict fire, smoke, and suppressant behavior in low gravity environments. The overall objective of this project is the development of an engineering tool that accurately predicts suppression agent dispersion and fire extinction for normal and low-gravity environments. To fulfill the overall objective, two computational tools, the Fire Dynamics Simulator (FDS) and a traditional commercial computational fluid dynamics software package (FLUENT) will be evaluated to determine if modeling of the suppression of fires can be validated for both normal and micro-gravity conditions. Experiments have been conducted to provide concentration and velocity data on suppression in normal gravity environments. The data is used to validate the computer modeling codes for the prediction of suppressant movement. Additionally, a model for local flame extinction based on a critical Damköhler number is developed for the prediction of suppression in both normal and low gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall benefit of this research study will be the development of a computational tool that can be used to determine the optimum placement and use of suppression systems within any three-dimensional geometry. The immediate impact of this study will be through enhancing the total fire safety aboard the International Space Station, Crew Exploration Vehicle, and other space vessels. A secondary impact of this tool will be an optimization of suppression system placement that will allow a reduction of the amount of agent needed to extinguish fires, thus reducing the weight of the agent that must be transported to and stored on the space vessel. The use of the computational tool will not be limited solely to space vessels. Direct application of the computational tool in Earth-based facilities will have merit through the potential optimization of localized suppression systems to prevent the unacceptable spread of fire and resulting destruction. This computational tool will have the most value in facilities containing equipment that mimics the electronics and component-rich configurations of spacecraft. The Earth-based facilities most likely to benefit from an analysis using this computational tool will be telecommunication switching stations, telecommunication utility rooms, electronics utility rooms, electronics manufacturing facilities, computer networking facilities, and other such telecommunication, electronics, or computer facilities.
Task Progress:	Through the 2006 work year, a series of experiments were conducted for validation purposes. To facilitate the validation experiments, the ground-based test mock-up construction was completed. The mock-up consisted of a reduced-scale simulated electronics rack that was one eighth of the volume of a typical rack found aboard a space vehicle such as the ISS. The dimensions of the mock-up were 0.5 meter by 0.5 meter by 1.0 meter. The ground-based rig was designed to allow for the addition of internal obstructions to increase the complexity of the geometry. These obstructions consisted of two parallel plates, where the separation distance between the two plates could be adjusted to various widths. This mock-up design allowed for a challenging geometrical configuration for fluid flow and provided a systematic test for model validation. A gaseous agent, Nitrogen, mixed with a small amount of tracer gas, NO2, was introduced into the test apparatus through a mounted nozzle located in the center of one of the four sides of the rig. The nozzle was an exact replica of the International Space Station (ISS) CO2 extinguisher nozzle. Data acquisition for the test apparatus over the 2006 work year consisted of agent concentration measurements. The concentration measurements were obtained in three distinct locations within the experimental apparatus during cold flow conditions (i.e., no flame and no fuel) using Planar Laser Induced Fluorescence (PLIF) of the tracer gas. By measuring the concentration level of the tracer gas, the concentrations of the suppression agent and local oxygen concentrations could be determined and compared with modeling predictions. The test setup utilized a Nd:YAG laser beam that was turned into a sheet projected through the channel created by the parallel plates. The position and direction of the sheet could be easily changed, as could the channel gap distance and the direction of the gap (horizontal or vertical). Because of the relatively weak fluorescence of NO2, an image intensifier, including a filter to reject the incident laser light, was used. Through precise gating, the intensifier ignored the incident laser pulse but observed the fluorescence of the NO2, which has a significantly longer lifetime than the incident light. A virtual representation of the ground-based test rig was composed using the Fire Dynamics Simulator (FDS) developed by the National Institute of Standards and Technology (NIST-USA). Modeling of the experimental data with the FDS code was used to determine if this code was appropriate for accurately modeling the suppression dispersion aspect of the problem. The FDS model of the ground-based test rig was constructed to scale with the identical dimensions and layout. A layer of fluid cells was added around the rig with open pressure boundaries to ensure that the internal pressure of the model did not rise unrealistically. All obstructions present in the actual ground-based rig were accounted for and included in the model. Additionally, four vents were included in the model that represented typical vents found in space vehicle electronics racks. Overall, the model contained 518,400 cells. These cells were approximately 1.0 cm (0.4 in.) in length in the y and z directions, and 0.635 cm (0.25 in.) in the x direction to ensure accurate separation of the parallel plates so as to match the data taken via experimentation. As mentioned above, the PLIF measurements allowed for obtaining the concentration of the tracer gas at locations between the parallel plates located within the ground-based test rig. Three test separations were chosen in an effort to determine a minimum spacing where entrance of the gas into the parallel plates would become more difficult. Due to constraints imposed by backscattering of the laser from the acrylic plates, 1.27 cm (0.5 in.) was found to be the smallest separation gap that still provided meaningful data. As such, in addition to the 1.27 cm (0.5 in.) gap, spacings of 1.91 cm (0.75 in.) and 3.18 cm (1.25 in.) were also examined in an effort to obtain concentration data. As mentioned above, gas was injected into the test rig during testing via a nozzle that was identical to the CO2 extinguisher nozzles located aboard the ISS. This nozzle was designed with eight outlets surrounding the tip of the nozzle in a radial fashion. A ninth outlet was located directly at the tip of the nozzle. In the PLIF experiments, the testing agent was composed of a mixture of 1000 ppm NO2 mixed with a balance of N2. The test mixture was introduced into

	the rig at a pressure of 2.4 MPa (350 psi) over a blowdown period of approximately five seconds. After the blowdown period, concentration data was recorded for an additional 15 seconds. Results were obtained as images recorded by a digital camera with appropriate lens filters and were processed using a MATLAB routine. A range of concentration values was obtained experimentally at the three separations tested. The peak concentrations ranged from 280 ppm of NO2 to 325 ppm of NO2. The smaller value was obtained at the 1.27 cm (0.5 in.) plate separation, and the larger value was obtained at the 3.18 cm (1.25 in.) plate separation. Predictions from the FDS modeling resulted in a fairly uniform concentration value of 400 ppm of NO2. This peak value did not vary greatly among the three plate separations. Comparison of the modeling results with the experimental results clearly shows marked differences. The modeling results produced by FDS overpredict the maximum concentration by approximately 120 ppm for the 1.27 cm (0.5 in.) case and by 75 ppm for the 3.18 cm (1.25 in.) case. The overall trends of the concentration histories are fairly similar, yet proper transport from the nozzle into the plates is not captured accurately. Even with these differences, the FDS model only overpredicts the peak NO2 concentrations at the middle plate location by 35% at the 1.27 cm (0.5 in.) plate spacing and by 20% at the 3.18 cm (1.25 in.) plate spacing. If an average of the concentration is taken after the blowdown period has occurred, these predicted concentrations improve slightly to 30% overprediction for the 1.27 cm (0.5 in.) spacing and 18% overprediction for the 3.18 cm (1.25 in.) spacing. While these differences are significant, the results from the model still have the potential to be useful in examining the transport aspect of this problem. Several issues will be explored in the 2007 work year in an effort to reduce the errors in the model predictions. First, a complete grid sensitivity analysis will be conducted to determine the effects of the grid. Second, a more detailed accounting of the difference between the ambient temperatures in the test rig and the inlet gases will be conducted. Within FDS, small differences in density due to temperature differences can have a significant impact. Third, the Large Eddy Simulation (LES) routine used to obtain results to date may not capture transport at an appropriate level of detail. An as alternative, a Direct Numerical Simulation (DNS) solver is available and will be explored to determine if improved results can be obtained. Finally, a commercial computational fluid dynamics modeling package (FLUENT) will be used to analyze the distribution aspect of the problem in an effort to obtain more reliable results.
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